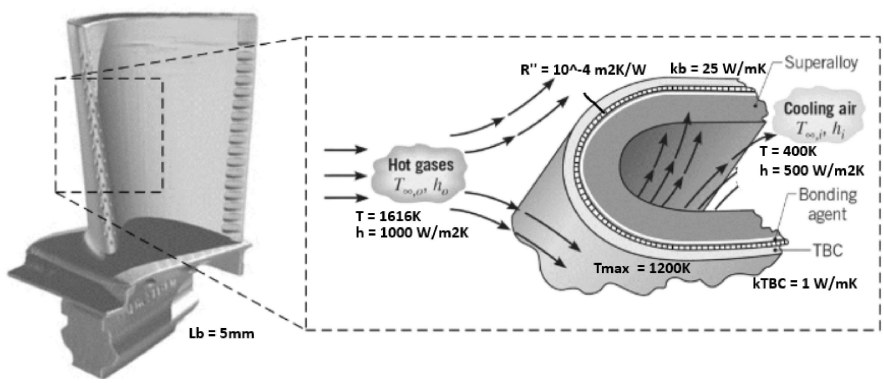
Problem 3

KNOWN. Thermal conductivity of blade alloy and ceramic, blade thickness and maximum allowable temperature, thermal resistivity of bonding agent, heat transfer coefficients of gasses, and temperature of gasses

FIND. (a) Resistance network of alloy without TBC, exterior temperature of alloy, check that it is below the maximum value; (b) Resistance network with TBC, exterior temperature of alloy, check that it is below maximum value

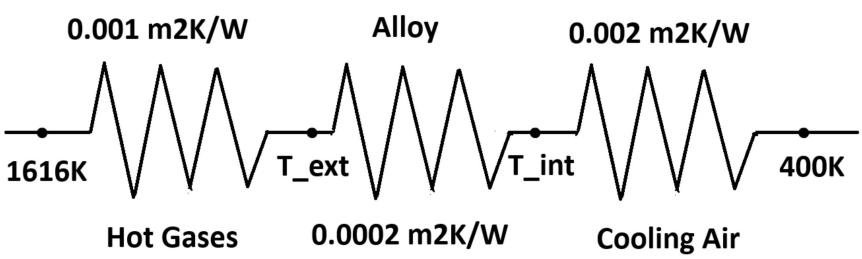
SCHEMATICS.



ASSUMPTIONS. Steady-state, radiation can be neglected, 1d heat conduction, thickness of the bonding agent is negligible

ANALYSIS

(a) Thermal resistance network



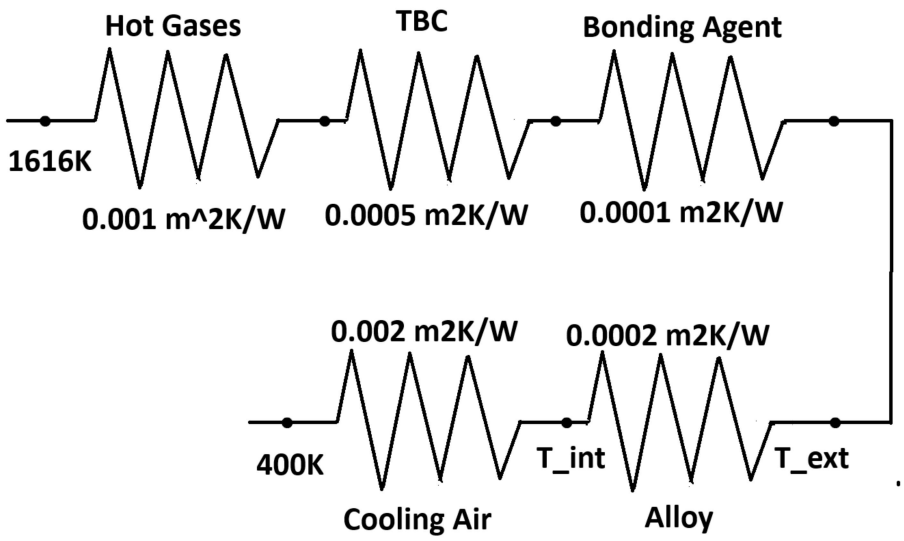
Temperature difference: $\Delta T = 1616K - 400K = 1216K$

Total thermal resistivity: $R''_{tot} = 0.0032 \frac{m^2 K}{W}$

$T_{ext} = 1616K - \frac{5}{16} \times 1216K = 1236K$

This is above the maximum allowable temperature for the alloy. Bad!

(b) Thermal resistance network



Same temperature difference of $\Delta T = 1216K$.

Total thermal resistivity: $R''_{tot} = 0.0038 \frac{m^2 K}{W}$

$T_{ext} = 1616K - \frac{16}{38} \times 1216K = 1104K$

This is below the maximum allowable temperature for the alloy. This is good! Since this is steady-state, the temperature will not change (assuming it is not acted upon in any way). The alloy will remain at a safe temperature.